

Lab 10: CMOS Inverter

ECE 0102: Microelectronic Circuits + Lab

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Team Roles:

Activity	Student Name
Circuit Construction	Stefan Zhang
Data Collection	Knight Liu
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1) Objectives

The objective of this lab is to construct a CMOS inverter using a CD4007UB transistor array and measure its voltage transfer characteristic [1]. Based on the measured V_{IN} - V_{OUT} data, the voltage transfer curve is plotted and the input threshold voltages V_{IL} and V_{IH} are determined.

2) Equipment Used

Table 1: Equipment used in Lab 10

Item	Quantity
CD4007UB CMOS transistor array	1
DC power supply	1 set
Digital multimeter	1
Breadboard and wires	1 set
Computer software for plotting	1

3) Action 1: CMOS Inverter Construction and Measurement

In this action, a CMOS inverter was constructed using one NMOS transistor and one PMOS transistor from the CD4007UB chip. The supply voltage was set to

$$V_{DD} = 5 \text{ V}.$$

The input voltage V_{IN} was applied to the common gate terminal of the NMOS and PMOS transistors. The output voltage V_{OUT} was measured at the common drain node. The NMOS source was connected to ground, and the PMOS source was connected to $+5 \text{ V}$.

For a CMOS inverter, when V_{IN} is low [2], the PMOS transistor is on and the NMOS transistor is off. Therefore, V_{OUT} is close to V_{DD} . When V_{IN} is high, the NMOS transistor is on and the PMOS transistor is off. Therefore, V_{OUT} is close to 0 V . In the transition region, both transistors conduct and V_{OUT} drops rapidly.

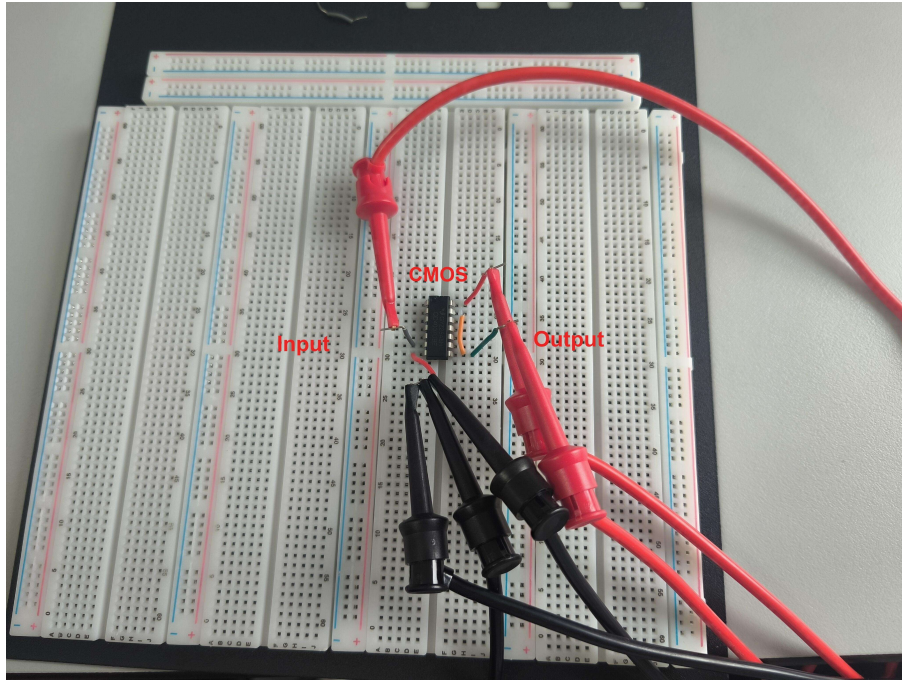


Figure 1: Circuit and equipment setup for the CMOS inverter measurement

The measured output voltage values are shown in Table 2. The third column was calculated using

$$\frac{\Delta V_{OUT}}{\Delta V_{IN}} = \frac{V_{OUT,i+1} - V_{OUT,i}}{V_{IN,i+1} - V_{IN,i}}.$$

Table 2: Measured voltage transfer data for Action 1

V_{IN} [V]	V_{OUT} [V]	$\Delta V_{OUT}/\Delta V_{IN}$
0.00	5.066	-0.12
0.50	5.006	-0.05
1.00	4.981	-0.60
1.50	4.680	-1.00
1.55	4.630	-1.06
1.60	4.577	-1.14
1.65	4.520	-1.24
1.70	4.458	-1.30
1.75	4.393	-1.46
1.80	4.320	-1.44
1.85	4.248	-1.84
1.90	4.156	-1.96
1.95	4.058	-2.26
2.00	3.945	-2.68
2.05	3.811	-3.48
2.10	3.637	-4.98
2.15	3.388	-8.26
2.20	2.975	-11.82
2.25	2.384	-12.46
2.30	1.761	-10.26
2.35	1.248	-5.56
2.40	0.970	-2.92
2.45	0.824	-1.96
2.50	0.726	-1.48
2.55	0.652	-1.24
2.60	0.590	-1.06
2.65	0.537	-0.92
2.70	0.491	-0.84
2.75	0.449	-0.78
2.80	0.410	-0.66
2.85	0.377	-0.66
2.90	0.344	-0.56
2.95	0.316	-0.58
3.00	0.287	-0.40
3.50	0.087	-0.17
4.00	0.004	-0.01
4.50	0.000	0.00
5.00	0.000	–

4) Action 2: Voltage Transfer Curve

Based on the measured data from Action 1, the voltage transfer curve of the CMOS inverter was plotted. The curve shows that the output voltage remains close to 5 V when the input voltage is low. As V_{IN} increases into the transition region, V_{OUT} decreases rapidly. When V_{IN} becomes high, the output voltage approaches 0 V.

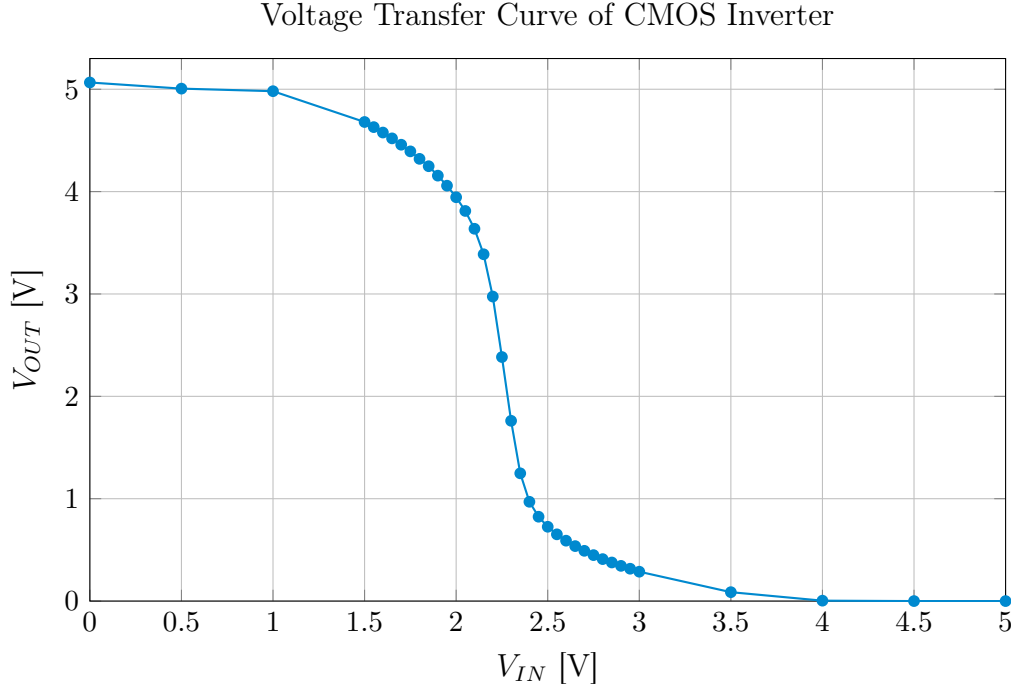


Figure 2: Measured voltage transfer curve of the CMOS inverter

The measured curve has the expected inverting behavior. At $V_{IN} = 0$ V, the output voltage is approximately 5.066 V, which corresponds to a logic high output. At $V_{IN} = 5$ V, the output voltage is approximately 0 V, which corresponds to a logic low output. The sharp transition occurs around $V_{IN} = 2.2$ V to 2.35 V.

5) Action 3: Determination of V_{IL} and V_{IH}

For a CMOS inverter, V_{IL} and V_{IH} are determined from the voltage transfer characteristic using the slope condition

$$\frac{dV_{OUT}}{dV_{IN}} = -1.$$

Since the measured data are discrete, the derivative was approximated by the finite difference

$$\frac{dV_{OUT}}{dV_{IN}} \approx \frac{\Delta V_{OUT}}{\Delta V_{IN}}.$$

Therefore, V_{IL} and V_{IH} were found from the two regions where the calculated slope is close to -1 .

From Table 2, the first point where the slope is close to -1 occurs at

$$V_{IN} = 1.50 \text{ V},$$

where

$$\frac{\Delta V_{OUT}}{\Delta V_{IN}} \approx -1.00.$$

Thus,

$$V_{IL} \approx 1.50 \text{ V}.$$

On the right side of the transition region, the slope changes from -1.06 at $V_{IN} = 2.60 \text{ V}$ to -0.92 at $V_{IN} = 2.65 \text{ V}$. Since -1 lies between these two values, linear interpolation was used:

$$V_{IH} = 2.60 + \frac{(-1) - (-1.06)}{(-0.92) - (-1.06)}(2.65 - 2.60).$$

Therefore,

$$V_{IH} = 2.60 + \frac{0.06}{0.14}(0.05) = 2.621 \text{ V}.$$

Thus,

$$V_{IH} \approx 2.62 \text{ V}.$$

Table 3: Input threshold voltages of the CMOS inverter

Parameter	Value
V_{IL}	1.50 V
V_{IH}	2.62 V

6) Discussion

The measured voltage transfer curve clearly shows the behavior of a CMOS inverter. When V_{IN} is low, the PMOS transistor conducts and pulls the output node toward V_{DD} , while the NMOS transistor is off. Therefore, the measured V_{OUT} stays close to 5 V. When V_{IN} is high, the NMOS transistor conducts and pulls the output node toward ground, while the PMOS transistor is off. Therefore, the measured V_{OUT} approaches 0 V.

The transition region appears between approximately 2.0 V and 2.6 V. In this region, both the NMOS and PMOS transistors conduct at the same time, so the output voltage changes rapidly with the input voltage. This is why the magnitude of $\Delta V_{OUT}/\Delta V_{IN}$ becomes large in the middle of the voltage transfer curve.

The maximum negative slope occurs around $V_{IN} = 2.25 \text{ V}$ to 2.30 V , where the calculated slope reaches approximately -12 . This indicates that a small change in input voltage produces a large change in output voltage. This steep transition is desirable for digital logic because it makes the inverter more resistant to small input noise.

The values of V_{IL} and V_{IH} were found using the condition

$$\frac{dV_{OUT}}{dV_{IN}} = -1.$$

There are several possible sources of measurement error. First, the input voltage was adjusted manually, so the actual V_{IN} may not be exactly equal to the nominal value listed in the table. Second, the CD4007UB transistors are not perfectly matched, so the switching point may not be exactly $V_{DD}/2$. Third, wire resistance, breadboard contact resistance, and multimeter resolution can also affect the measured values slightly. However, the overall trend of the measured voltage transfer curve is consistent with the expected CMOS inverter behavior.

7) Conclusion

In this lab, a CMOS inverter was constructed using the CD4007UB chip [3]. The input voltage V_{IN} was swept from 0 V to 5 V, and the corresponding output voltage V_{OUT} was measured. The measured voltage transfer curve confirms the expected inverting behavior of the CMOS inverter.

When V_{IN} is low, V_{OUT} is close to 5 V. When V_{IN} is high, V_{OUT} is close to 0 V. The output voltage changes sharply in the transition region, which occurs near the middle of the supply voltage range.

Using the finite-difference approximation of the slope, V_{IL} and V_{IH} were determined from the points where the slope of the voltage transfer curve is approximately -1 . The measured values are

$$V_{IL} \approx 1.50 \text{ V}$$

and

$$V_{IH} \approx 2.62 \text{ V}.$$

These values describe the input voltage boundaries for reliable low and high logic recognition in the measured CMOS inverter.

References

- [1] SCUPI 2026S Microelectronics Circuits Lab Manual, Lab #10: CMOS Inverter.
- [2] Sedra, A. S., Smith, K. C., Carusone, T. C., and Gaudet, V., *Microelectronic Circuits*, Oxford University Press.
- [3] Texas Instruments, *CD4007UB CMOS Dual Complementary Pair Plus Inverter Datasheet*.